

Daris Chen

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EDUCATION

University of California San Diego <i>Bachelor of Science (B.S.) in Computer Engineering</i>	March 2026 <i>San Diego, CA</i>
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WORK EXPERIENCE

Software Engineer Intern - AI/ML (Lead Engineer, PatentIQ) <i>Product Manager Accelerator</i>	Jan 2026 – Present <i>Remote</i>
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- Constructed an AI-powered patent search and drafting assistant hosted on Vercel, EC2, and Supabase using Next.js, Auth0, and FastAPI to simplify intellectual property research for early-stage founders.
- Programmed a hybrid semantic search engine combining OpenAI text-embeddings and PostgreSQL pgvector to perform high-dimensional similarity ranking and recommendation system over global USPTO patent datasets.
- Orchestrated Agile development workflows via ClickUp for a 14 member cross functional team while serving as lead architect and primary contributor to deliver the foundational PatentIQ AI technical MVP.
- Built recursive SQL query builder with Zod-validated JSON filters and implemented an LRU cache to reduce latency.

Lead Full-Stack Engineer <i>Groundwork Books</i>	Aug 2025 – Mar 2026 <i>San Diego, CA</i>
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- Led architectural migration of production e-commerce to Next.js and Square for live inventory synchronization.
- Engineered a semantic search layer using Pinecone with integrated text embeddings, enabling topic and concept-based discovery across 4,000+ SKUs without requiring exact title matches.
- Architected a two-tier caching system: a look-aside Redis reducing Square API latency by 90% (sub-200ms p99), and a client-side request coalescing hook batching 50+ concurrent inventory lookups into an API call with 2-minute TTL cache.

FishSense Artificial Intelligence Researcher <i>UCSD Engineers for Exploration (E4E)</i>	Feb 2025 – Mar 2026 <i>San Diego, CA</i>
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- Developed a custom R-CNN training pipeline (ResNet-50 backbone) for underwater laser-profiling, standardizing the Detectron2 codebase, reducing inference setup time by 40% for new researchers and increasing reproducibility.
- Devised a dual-laser photogrammetry algorithm to extract biological metrics (fish length/mass) from 2D video feeds, achieving sub-5% error rates in pixel-to-centimeter conversion.

PROJECTS

FlipperZillow <i>Next.js, Python, PyTorch, AMD ROCm, Three.js, WebSpatial</i>	Feb 2026 – Present
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- Designed an AI house tour platform named FlipperZillow using Next.js and Claude for semantic search, piping DFormerV2 spatial analysis into Claude Sonnet 4.6 to generate property scripts and ElevenLabs audio narration.
- Integrated an AMD ROCm SSH pipeline to process Realtor.com property images cascading DepthAnythingV2 into DFormerV2 and extracting WebSpatial compatible 3D objects using META's SAM3D Object.
- Assembled a 3D experience combining Google Maps and WebSpatial for Vision Pro with a Three.js fallback.

Simplerfy <i>Next.js, Claude API, Chrome Extension API, LaTeX</i>	Jan 2025 – Feb 2025
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- Constructed a job application management platform with a browser extension using the Chrome Extension API to automate form-filling across 15+ job boards.
- Programmed an AI optimization engine using the Claude API to evaluate job descriptions, extracting unstructured data from PDF resumes to automate profile population and generate tailored LaTeX versions with scoring systems.

Personal Portfolio Site <i>React, TypeScript, Node.js, Express.js, Tailwind CSS, Vercel</i>	Oct 2024 – Present
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- Developed a custom SSG pipeline transforming raw Markdown/LaTeX into statically optimized HTML at build time, achieving perfect Lighthouse performance scores (100/100).
- Implemented dynamic open-graph image generation and edge-cached asset delivery, reducing First Contentful Paint (FCP) to under 0.8s globally.

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, JavaScript/TypeScript, HTML/CSS, ARM Assembly, SQL, Verilog/SystemVerilog, Go, Cython

Libraries & Frameworks: PyTorch, TensorFlow, CUDA, ROCm, cuDNN, Transformers, NumPy, pandas, Matplotlib, Scikit-learn, Detectron2, tqdm, React, Next.js, Node.js, Express.js, Three.js, Svelte, Tailwind CSS, Zod, DepthAnythingV2, DFormerV2, SAM3D Object, OpenAI, Claude, Signal Processing, FFT

Tools & Technologies: Git, GitHub Actions, Docker, Kubernetes, Linux, Clickup, Jira, Jest, Puppeteer, GCP, gRPC, EC2, GCC, OpenCL, Figma, VS Code, CodeMirror, Wandb, Jupyter, Hugging Face, Vercel, ESLint, Turbopack, OSM Nominatim, Open-Meteo, OpenStreetMap, Intel FPGA ModelSim, WebSpatial (Vision Pro), Auth0, SSH, HB100, RF Engineering, ESP32

Databases: PostgreSQL, MongoDB, Upstash Redis, Firebase, Pinecone, pgvector, Supabase